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Practical No 3

AIM : Implementation of flag manipulation and process control instructions

THEORY :

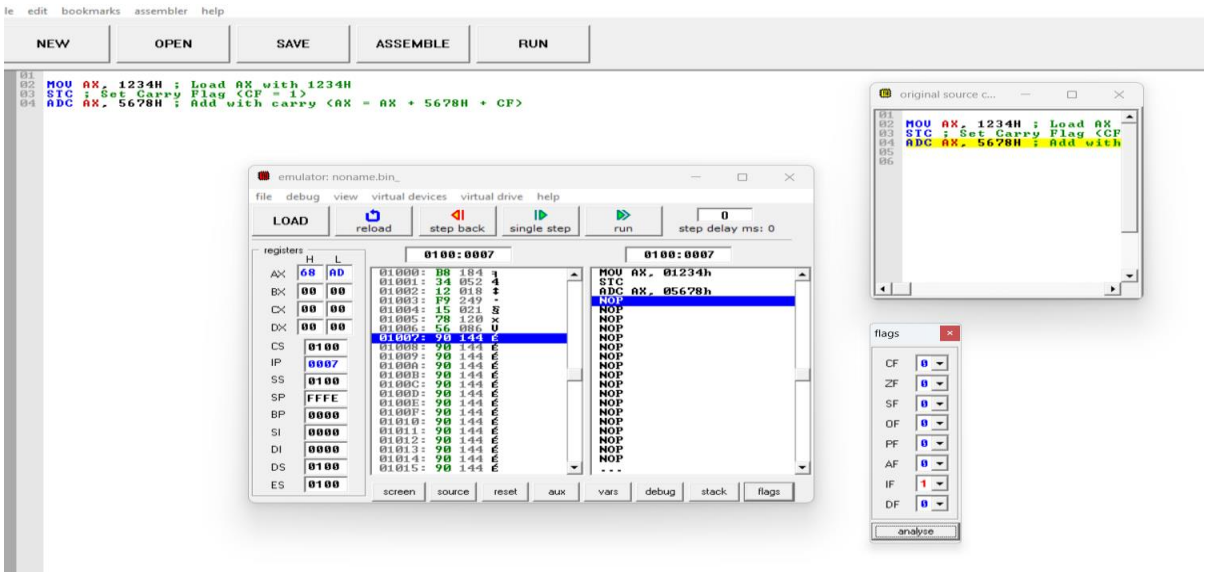
- **Flag manipulation instructions :** to control the flow of carry and direction flag
  - STC (Set carry flag)
  - CLC (clear carry flag)
  - CMC (complement carry flag)
  - CLD (clear direction flag)
  - STD (set direction flag)
- **Process control instructions :** used to control the flow of processes using various system instructions
  - HLT (halt)
  - o WAIT (wait)
  - o LOCK (bus lock)
  - o NOP (no operation)
  - o ESC (Escape)

PROGRAM :

- **Flag manipulation instructions:**

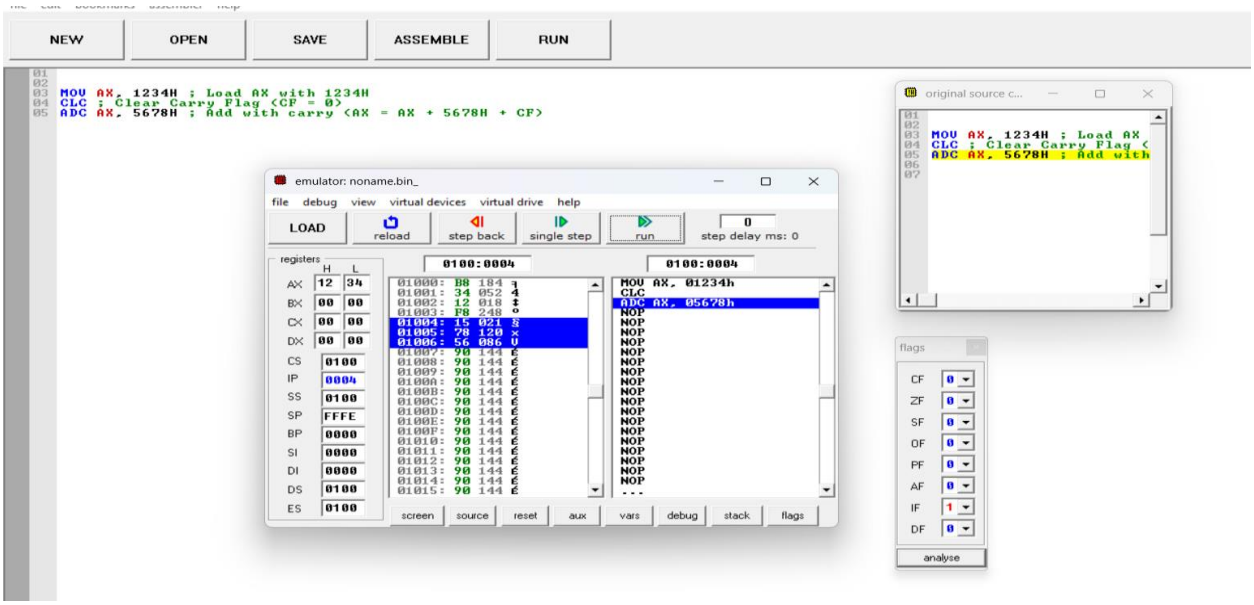
❖ **STC (Set Carry Flag)**

Purpose: Sets the Carry Flag (CF) = 1.



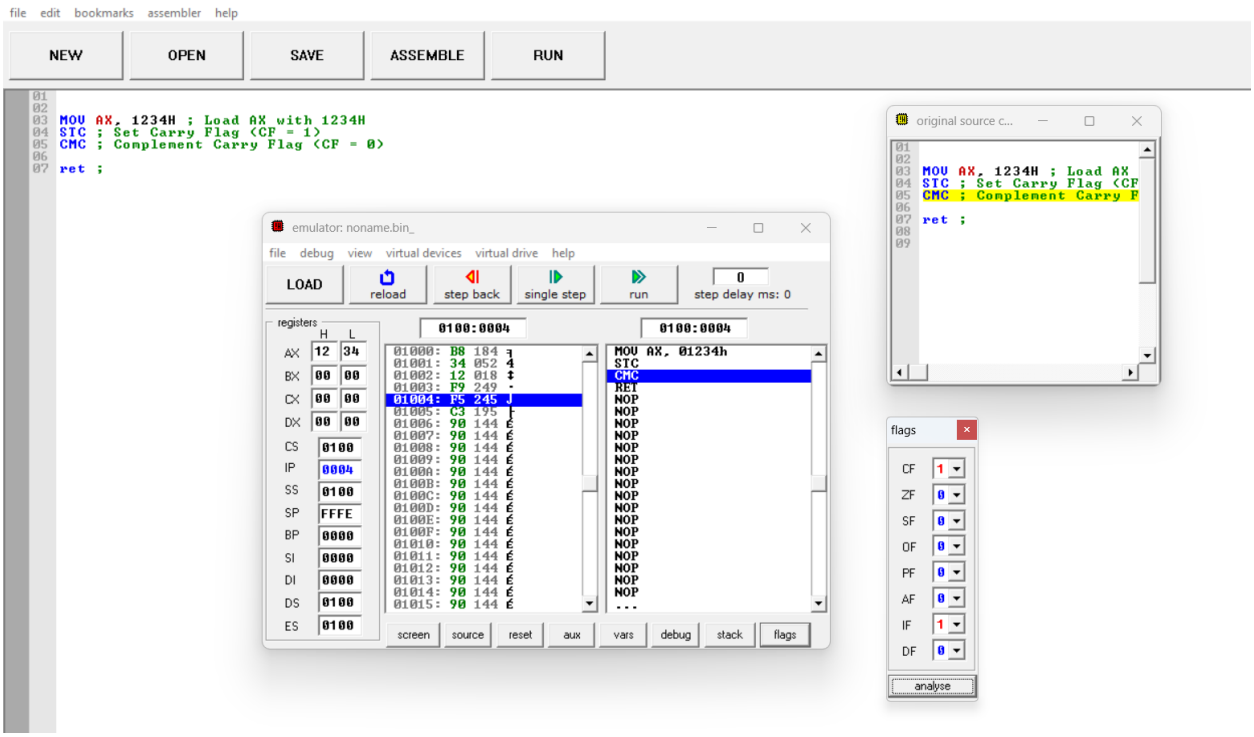
- **CLC (Clear Carry Flag)**

  Purpose: Clears the Carry Flag (CF) = 0.



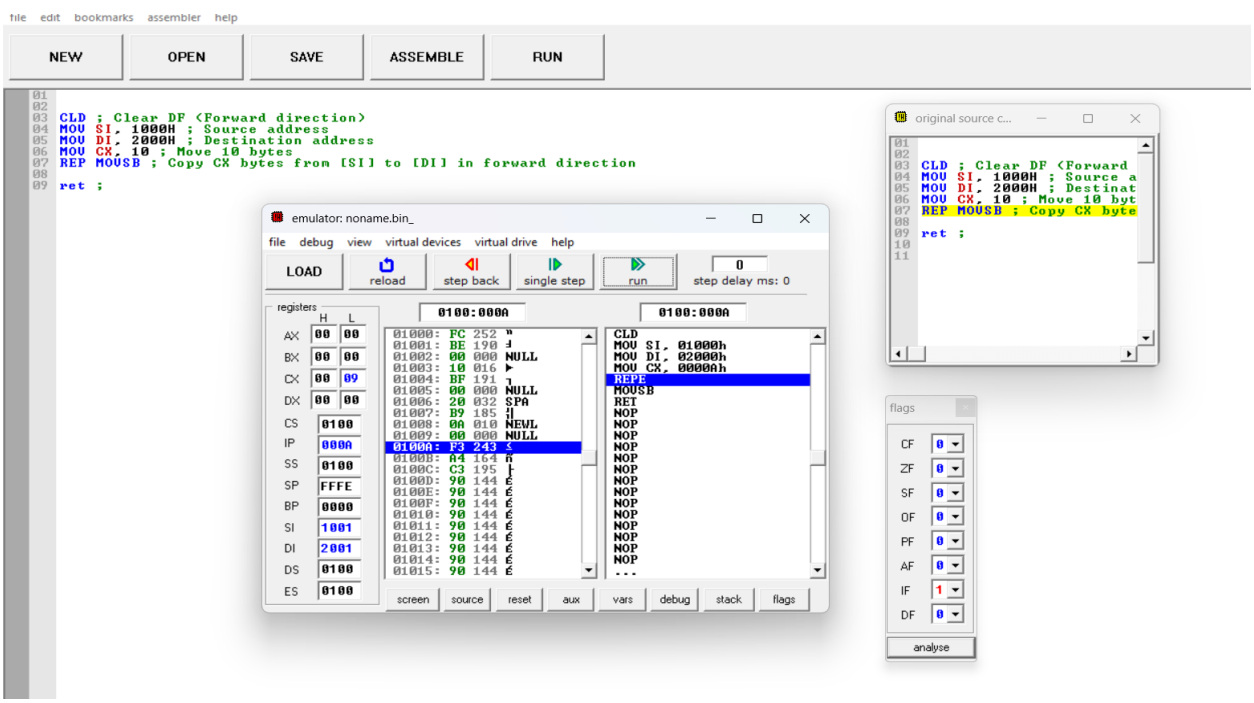
- **CMC (Complement Carry Flag)**

  Purpose: Toggles the Carry Flag (CF = 1 → 0 or CF = 0 → 1).



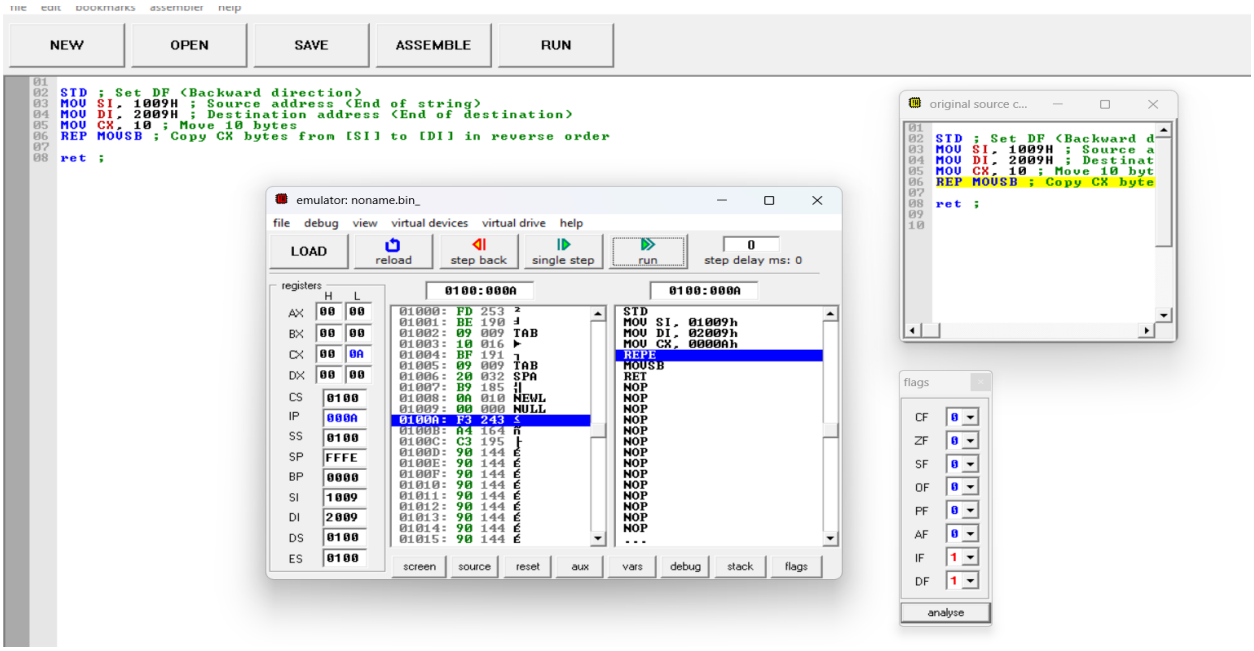
- **CLD (Clear Direction Flag)**

Purpose: Clears the Direction Flag (DF)



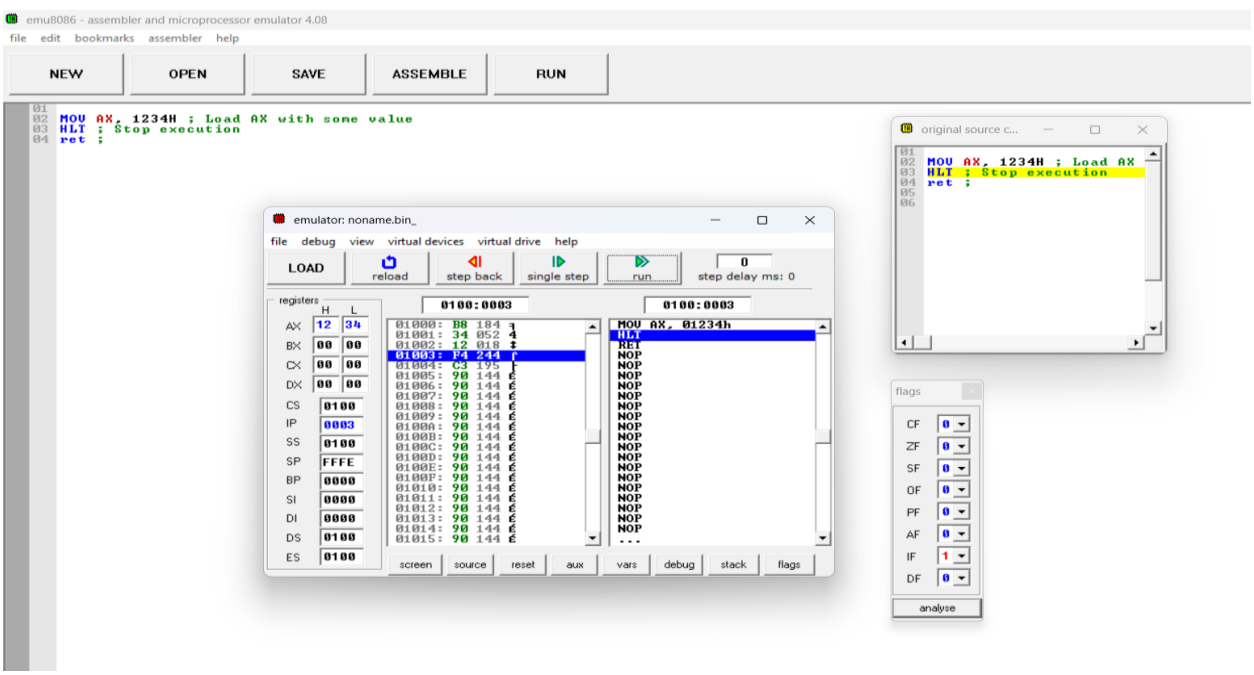
- **STD (Set Direction Flag)**

Purpose: Sets the Direction Flag (DF) = 1, making string operations process data from high to low memory.



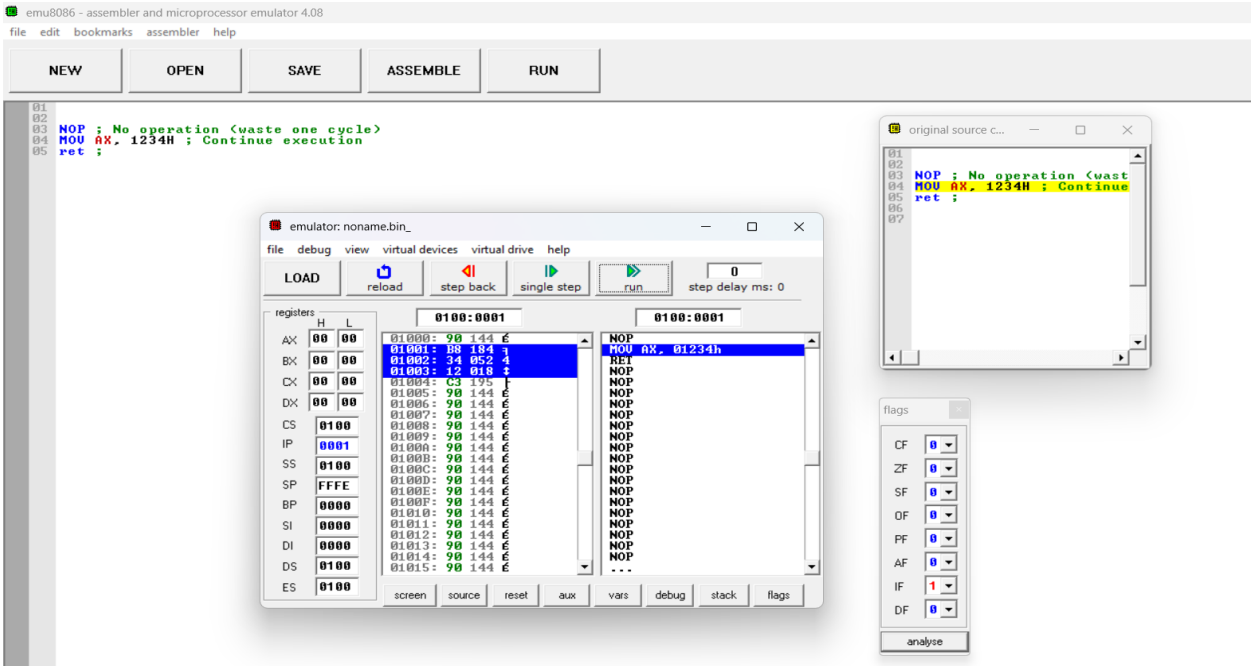
- **Process control instructions:**
- **HLT (Halt)**

  Purpose: Stops the processor until an interrupt or reset occurs.



- **NOP (No Operation)**

  Purpose: Does nothing for one clock cycle (used for delays or timing adjustments).



- **WAIT (Wait for Test Pin)**

  Purpose: Pauses execution until the Test (TST) pin is cleared.

  WAIT ; Processor waits until external hardware clears TEST pin

  MOV AX, BX ; Execution resumes after WAIT

- **LOCK (Bus Lock)**

  Purpose: Ensures exclusive access to shared resources in multiprocessor systems.

  MOV AX, 1234H

  LOCK ADD [5000H], AX ; Add AX to memory at 5000H, locking bus access

- ESC (Escape to Coprocessor)

  Purpose: Sends an Escape instruction to an external coprocessor (like 8087 NDP).

  ESC 0, [1000H] ; Send escape instruction to coprocessor at memory location 1000H

CONCLUSION:

We demonstrated how each flag manipulation and process control instruction works in 8086

Assembly Language.